Empirical Validation of Tensor Logic: Unifying Symbolic and Neural Reasoning

Swapn Shah
School of Data Science
University of North Carolina at Charlotte
sshah100@charlotte.edu

Advised by: Dr. Wlodek Zadrozny
Department of Computer Science

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Abstract

The unification of symbolic reasoning and neural networks remains a central challenge in artificial intelligence. Symbolic systems offer reliability and interpretability but lack scalability, while neural networks provide learning capabilities but sacrifice transparency. Tensor Logic, proposed by Domingos, suggests that logical rules and Einstein summation are mathematically equivalent. This paper provides empirical validation through two experiments: computing transitive closure on a genealogy graph with 1,972 individuals, and training a neural network for compositional inference that achieves 100% accuracy on zero-shot multi-hop queries.

1 Introduction

Artificial intelligence research has historically followed two divergent paths. The symbolic tradition, rooted in mathematical logic and exemplified by systems like Prolog and Datalog, emphasizes precise reasoning over structured knowledge representations. These systems excel at logical inference and explanation generation but struggle with noisy real-world data and face scalability limitations.

The connectionist tradition, dominated by deep learning, takes the opposite approach. Neural networks learn distributed representations from raw data and have achieved remarkable success in perception tasks. Yet these systems operate as black boxes and struggle with systematic generalization, often failing on compositional tasks that require combining learned concepts in novel ways.

Recent approaches have sought tighter integration, embedding logical constraints into neural architectures or learning logical rules through differentiable relaxations. Despite progress, most neuro-symbolic systems still treat neural and symbolic components as separate modules that must be carefully interfaced.

2 Problem Statement and Motivation

Domingos proposes Tensor Logic as a solution to the neuro-symbolic integration problem. The key insight is that Datalog rules and Einstein summation are fundamentally the same operation,

differing only in their atomic data types. A logical rule performs joins and projections over Boolean tensors, while a neural network performs the same operations over real-valued tensors.

However, this theoretical framework lacks comprehensive empirical validation. The central research question is: *Can tensor operations truly perform both exact symbolic reasoning and continuous neural inference while maintaining logical consistency and compositional generalization?*

This question is important for several reasons. First, if validated, Tensor Logic would provide a principled foundation for building AI systems that combine strengths of both paradigms. Second, it would enable reasoning systems to leverage GPU acceleration. Third, it would provide a path toward AI systems that can both learn from data and reason with learned knowledge in a unified way.

2.1 Theoretical Foundation

Consider the ancestry rule in Datalog:

$$\text{Ancestor}(x, z) \leftarrow \text{Ancestor}(x, y) \wedge \text{Parent}(y, z) \tag{1}$$

Let A and P be Boolean matrices representing the Ancestor and Parent relations. The rule can be written as:

$$A_{xz} = H\left(\sum_y A_{xy} \cdot P_{yz}\right) = H(A \times P) \tag{2}$$

where H is the Heaviside step function. For embedding space reasoning, relations are transformation matrices $M_r \in \mathbb{R}^{d \times d}$. For compositional queries:

$$v_{\text{pred}} = v_{\text{subj}} \times M_{r_1} \times M_{r_2} \times \cdots \times M_{r_n} \tag{3}$$

3 Methods

3.1 Experiment 1: Symbolic Transitive Closure

Dataset. We used the Bible-Data repository, which provides structured genealogical data. After filtering for parent-child relationships and normalizing edge directions, we obtained a directed graph with $N = 1{,}972$ nodes and $E = 1{,}727$ edges.

Algorithm. The Parent relation is represented as a Boolean matrix $P \in \{0, 1\}^{N \times N}$ where $P_{ij} = 1$ if person i is a parent of person j. We implement forward chaining through iterative matrix multiplication. The Ancestor matrix is initialized to $A^{(0)} = P$. Each iteration applies:

$$A^{(t+1)} = H\left(A^{(t)} + A^{(t)} \times P\right) \tag{4}$$

In NumPy: `new = np.einsum('xy,yz->xz', Ancestor, Parent)` followed by `Ancestor = np.where((Ancestor + new) > 0, 1, 0)`. The algorithm terminates when no new edges are discovered.

Verification. We verify correctness through three checks: (1) Containment: $P \subseteq A$, (2) Closure: $A \times P$ adds no new edges, (3) Acyclicity: $\text{diag}(A) = 0$.

3.2 Experiment 2: Embedding Space Reasoning

Dataset. We used the Countries dataset, extracting two relations: *is_capital_of* (245 pairs) and *is_located_in* (245 pairs), yielding 489 unique entities and 490 training facts. The training data contains no direct capital-continent links.

Model Architecture. We implement learnable entity embeddings $E \in \mathbb{R}^{489\times 64}$ and relation matrices $M \in \mathbb{R}^{2\times 64\times 64}$, both with Xavier initialization. The forward pass computes:

$$v_{\text{pred}} = \text{einsum}(\texttt{'bi,bij->bj'}, E[s], M[r]) \tag{5}$$

Scoring uses dot product: scores $= v_{\text{pred}} \times E^T$.

Training. We use cross-entropy loss with Adam optimizer (learning rate 0.005, 500 epochs, full-batch):

$$L = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(\text{score}_{i,y_i})}{\sum_{j=1}^{489}\exp(\text{score}_{i,j})} \tag{6}$$

Zero-Shot Inference. At test time, we evaluate queries requiring chained relations. Given a capital, we predict its continent by composing learned matrices:

$$v_{\text{continent}} = v_{\text{city}} \times M_{\text{capital}} \times M_{\text{located}} \tag{7}$$

4 Results

4.1 Experiment 1: Symbolic Transitive Closure

The iterative tensor approach successfully computed the complete transitive closure. Table 1 summarizes key statistics.

Table 1: Transitive Closure Statistics

Metric	Value
Initial nodes	1,972
Initial edges	1,727
Convergence iteration	74
Final edges	33,945
New edges discovered	32,218

The algorithm required 74 iterations to reach fixpoint, aligning with expected generational spans. All three correctness checks passed. We examined specific lineages: Adam (821 descendants, 0 ancestors, first generation: Abel, Cain, Seth) and Abraham (698 descendants, 20 ancestors, with ancestor chain matching Genesis 11).

4.2 Experiment 2: Embedding Space Reasoning

The model converged rapidly. Initial loss was 6.19 (matching ln(489) for uniform random predictions), dropping to 0.003 by epoch 500. Table 2 shows zero-shot compositional inference results.

The model achieved 100% accuracy across all seven test cities, correctly inferring continents by composing learned transformation matrices. This validates that relation matrices learned during single-hop training compose correctly for multi-hop inference.

Table 2: Zero-Shot Compositional Inference Results

City	Prediction	Score	Correct
Tokyo	Asia	79.2	✓
Berlin	Europe	72.4	✓
Cairo	Africa	121.4	✓
Lima	Americas	79.1	✓
Canberra	Oceania	61.9	✓
New Delhi	Asia	65.4	✓
King Edward Point	Antarctic	71.0	✓

Hubness Analysis. In preliminary experiments with different random seeds, we observed failures from the hubness problem in high-dimensi–onal dot-product spaces. For one seed, "What continent is Tokyo in?" incorrectly predicted Antarctic (score 40.41, norm 2.33, cosine 0.616) instead of Asia (score 39.58, norm 2.22, cosine 0.633). Despite higher cosine similarity to Asia, Antarctic's larger embedding norm produced higher dot product score. This can be mitigated by normalizing embeddings to unit length.

5 Discussion

Our experiments validate two core claims from Tensor Logic. **Claim 1:** A Datalog rule is an einsum over Boolean tensors with a step function. Experiment 1 demonstrates this directly, the recursive transitive closure rule is exactly implemented as `np.einsum` with Heaviside function, discovering all 33,945 ancestor relationships from 1,727 parent links. **Claim 2:** Reasoning in embedding space uses forward chaining over learned transformation matrices. Experiment 2 shows relations learned as 64×64 matrices compose correctly, achieving 100% accuracy on zero-shot two-relation queries.

The framework offers practical advantages. Both einsum and matrix multiplication are highly optimized on GPUs, enabling reasoning systems to leverage hardware acceleration. The tensor operations are differentiable, enabling end-to-end training. Learned relation matrices can be composed at test time for zero-shot inference. The tensor representation provides interpretability through explicit matrix representations.

Limitations suggest future directions. Our experiments use relatively small datasets; scaling to web-scale graphs requires sparse representations. The hubness problem requires normalization strategies. Our compositional experiments involve two-relation chains; more complex queries with branching, negation, or aggregation would provide stronger tests. Learning logical structure itself would further demonstrate unification of symbolic and neural approaches.

6 Conclusion

We provided empirical validation of the Tensor Logic framework. Through two experiments, we demonstrated that logical inference can be performed through tensor operations in both discrete symbolic and continuous embedding spaces.

Experiment 1 showed recursive Datalog rules for transitive closure are exactly equivalent to iterative matrix multiplication with Heaviside function. Applied to genealogy data, the tensor-based algorithm correctly discovered all 33,945 ancestor relationships, converging in 74 iterations.

Experiment 2 showed relations can be learned as transformation matrices that compose correctly for multi-hop reasoning. A model trained only on single-hop facts achieved 100% accuracy on zero-shot compositional queries.

These results support Tensor Logic's central claim: that Datalog rules and Einstein summation are fundamentally equivalent operations. This equivalence provides a principled foundation for unifying symbolic reasoning and neural learning. Code and datasets are available at `https://github.com/sshah100-clt/tensor_logic_implementation`.

References

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Appendix A: AI Tool Disclosure

All experimental results, data analysis, mathematical derivations, research insights, and conclusions are original work. The core intellectual content, research methodology, experimental design, and interpretation of results were developed independently without AI assistance. AI tools were used only for formatting, clarification of existing concepts, routine coding tasks, and generating the LaTeX document structure. All AI-generated content was carefully reviewed and fact-checked.

Code Availability

The complete code and implementation for both experiments are publicly available at: `https://github.com/sshah100-clt/tensor_logic_implementation`